

Peel Park Aircraft Soundscape

Recording Information

Location	Peel Park, Salford, UK
Coordinates	53.487742° N, -2.269315° W
Date & Time	12 March 2025, 08:34 GMT
Recording Equipment	Zoom F8n / Røde NT-SF1 ambisonic microphone
Recording Format	4-channel B-format (AmbiX, ACN/SN3D), 24-bit, 96 kHz

Processing & Render Details

The original 4-channel B-format ambisonic recording was decoded using a first-order UHJ stereo rendering method, optimised for headphone presentation. A 38-second excerpt was processed in REAPER 7.35 using the AmbiX VST plugin suite (Kronlachner, 2014). Spatial decoding used the ambix_binaural_o1 plugin with the icosahedron-3h3v preset. A gain adjustment of +7.2 dB was applied via JS: Volume Adjustment, followed by limiting via JS: Master Limiter (ceiling -1.0 dBTP). Final metering used Youlean Loudness Meter 2. Output rendered at 24-bit, 44.1 kHz, 2-channel stereo WAV (2067 kbps), 16 April 2025.

Processing Chain

DAW	REAPER 7.35
Ambisonic decoder	AmbiX VST — ambix_binaural_o1, icosahedron-3h3v preset
Gain adjustment	JS: Volume Adjustment (+7.2 dB)
Limiter	JS: Master Limiter (ceiling -1.0 dBTP)
Loudness meter	Youlean Loudness Meter 2
Render date	16 April 2025

Output Specification

Format	WAV, 24-bit PCM
Sample rate	44,100 Hz
Channels	2 (stereo)
Bit rate	2067 kbps
Duration	0:38.000

Loudness Metrics

Parameter	Value
Integrated Loudness (LUFS-I)	-18.0 LUFS
Short-Term Loudness max (LUFS-S)	-14.6 LUFS
Momentary Loudness max (LUFS-M)	-11.2 LUFS
Loudness Range (LRA)	7.0 LU
True Peak	-1.3 dBTP
Clipping	0 occurrences

Loudness normalisation target: -18 LUFS ±0.5 (ITU-R BS.1770-4 / AES TD1004.1.15-10).

Rationale

Selected for its representation of dynamic soundscape conditions typical of urban park settings. The overhead aircraft contributes a momentary increase in eventfulness, offering contrast within the wider dataset and supporting evaluation of perceptual transitions. Peel Park is a consistent recording site in the study, enabling comparative interpretation across time and environmental conditions.

REAPER Render Analysis

REAPER Render Wed Apr 16 16:08:26 2025
WAV: 24bit PCM, 44100Hz, 2ch, 2067kbps

File	Length	Peak	Clips	max LUFS-M	max LUFS-S	LUFS-I	LRA
Peel Park Aircraft.wav	0:38.000	-1.3	0	-11.2	-14.6	-18.0	7.0

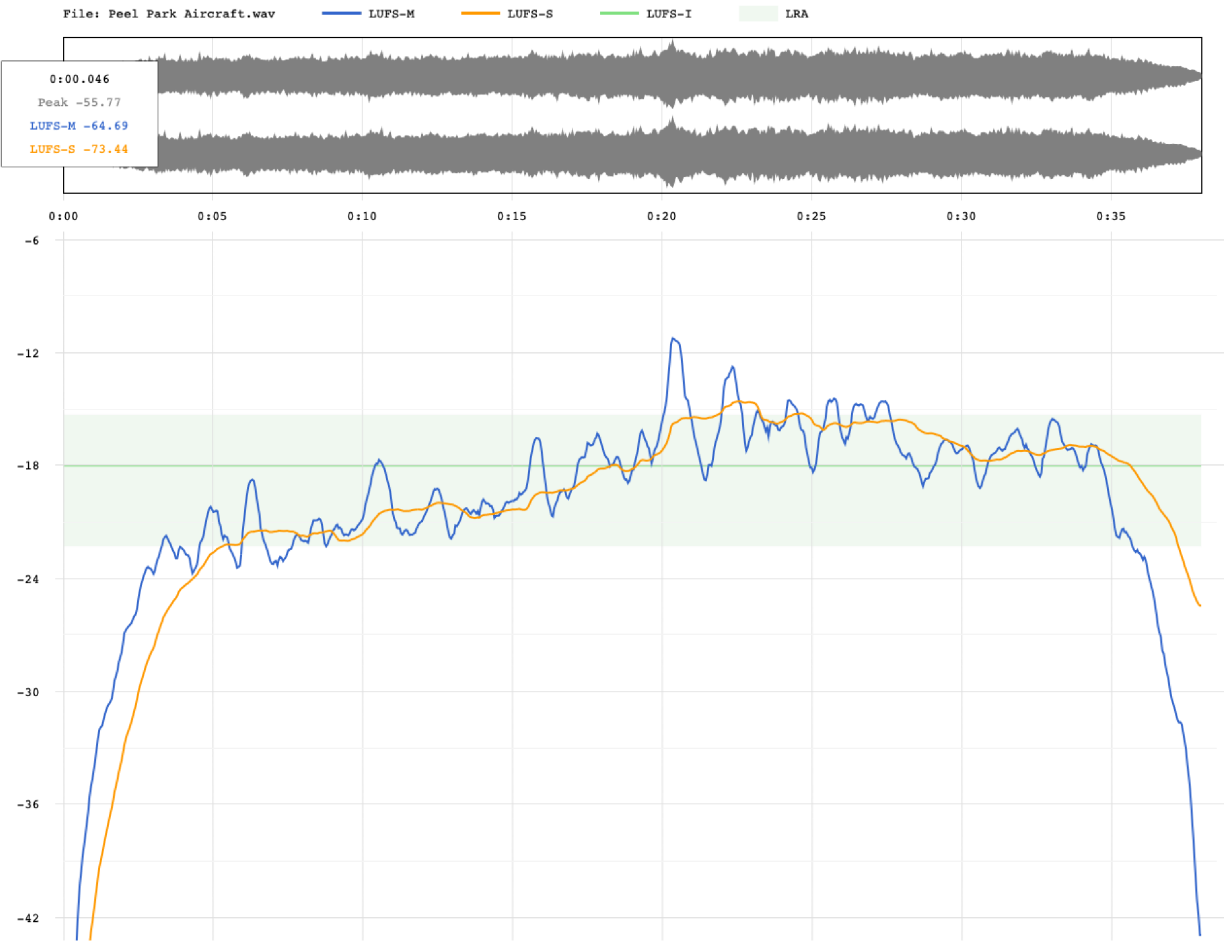


Figure: REAPER loudness render report — Peel Park Aircraft Soundscape. Waveform display (top) and integrated loudness profile (LUFS-M blue, LUFS-S orange, LUFS-I green, LRA shaded region).